

Federated Learning for Smart Agriculture: Challenges and Opportunities

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Abstract—FL is a machine learning approach that allows knowledge sharing with privacy maintenance and cost reduction. FL has the potential to revolutionize the smart agriculture sector by enabling farmers to train and deploy machine learning models on their own devices, without the need to share their data with a central server. In this article, we review the applications of FL in smart agriculture, such as disease diagnosis and severity assessment, the Internet of Agricultural Things (IoAT), and yield forecasting. We also highlight the challenges and opportunities of using FL for smart agriculture.

Index Terms—Smart Agriculture, Federated Learning, Distributed Learning, Disease Detection, Severity Assessment, Yield Forecasting.

I. INTRODUCTION

Smart agriculture refers to smart and efficient agriculture with the use of technologies such as the Internet of Things (IoT), sensors, location systems, robots, and artificial intelligence in farming practices [1]. The goal of smart agriculture is to improve the cost effectiveness and sustainability of agriculture while improving crop yields [2]. Machine learning (ML) techniques are used to predict crop yield, early detection of crop diseases, estimation of crop damage by weather parameters such as atmospheric temperature and humidity, assessment of disease severity, and detection of soil nutrition, water quality management, crop monitoring, livestock monitoring, etc. [1]. Researchers have developed various solutions based on artificial neural networks (ANN) to simplify agricultural tasks. In general, farm data is obtained from various sources such as IoT sensors, organizational data archives, and users and are then transferred to cloud servers where machine learning models are trained. This approach has a significant limitation, especially when the training data set

contains sensitive information, as it poses security threats, and this gives rise to concerns about the privacy of the data [3].

In 2016, Google introduced FL, an emerging approach that addresses this limitation by allowing knowledge sharing while maintaining privacy and reducing costs [4]. FL allows devices to collectively learn a shared model while keeping learning data stored on devices, thus separating machine learning functionality from the need for centralized data storage [5]. FL is also beneficial in resource-constrained situations, such as mobile devices, IoT devices, or networks with restricted bandwidth, because it tries to be both effective and scalable [6]. Figure 1 shows the general applications of FL in agriculture, advantages, challenges and limitations of using FL.

FL in smart agriculture has revolutionized data storage and analysis in the field. Using the power of distributed computing, FL allows researchers to train ML models using data from multiple sources, such as sensors, drones, and IoT devices, without the need to centralize the data. This decentralized approach not only ensures data privacy and security, but also enables real-time decision-making and optimization of farming practices. With FL, farmers can now benefit from the collective intelligence of a network of devices, resulting in more accurate predictions, improved crop yields, and sustainable agricultural practices without compromising data privacy [7]. This innovative technology has the potential to transform the agriculture industry by enabling efficient resource allocation, reducing environmental impact, and, ultimately, ensuring food security for a growing global population.

In an FL network, local nodes(end devices) send their updated weights to the server, which aggregates these weights and sends back the updated model weights for further training at the edge nodes. This process continues until convergence

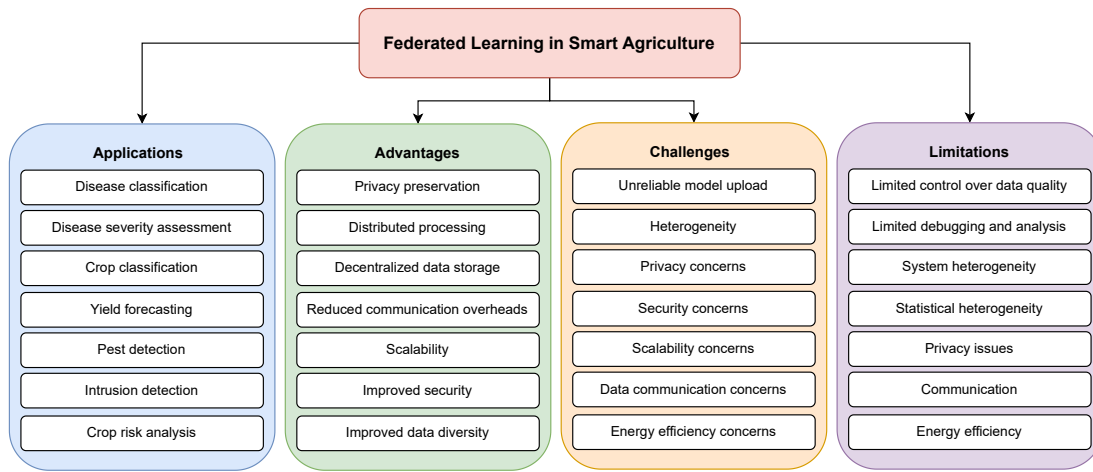


Fig. 1. Federated Learning in Smart Agriculture

is achieved [8]. The architecture of FL in smart agriculture is depicted in Figure 2. The following steps outline the process.

- 1) Various sensors deployed in farms monitor and collect data related to agriculture such as soil conditions, weather, irrigation, light, temperature, positioning, and air quality
- 2) The collected data is then transmitted to the local database
- 3) The local sites train their respective model using the data in local database
- 4) Updates of locally trained models are uploaded to the central server
- 5) The collected updates from multiple clients aggregated using aggregation algorithms such as FedAvg, FedProx
- 6) The updated model weights are sent back to the clients

A. Advantages of FL in smart agriculture

The utilization of ML in the field of agriculture offers numerous benefits, such as the ability to detect crop diseases at an early stage, detect intrusions, manage crops, and assess the situation and severity of various issues. However, the implementation of FL introduces additional advantages over ML, such as the preservation of privacy for crop data, reduction in communication costs, scalability, increased data diversity and many more [7]. Some of the advantages of FL in smart agriculture are described in the following subsections.

1) *Privacy Preservation*: In the traditional ML approaches, there is a concern about privacy as data is exchanged between clients and the server. However, FL addresses this issue by storing crop data on the device and only transmitting model updates to the central server. By keeping the data on the device, FL ensures the protection of raw data, reduces the chances of data leakage, and ensures secure operations. The FL model gathers crop data and stores it exclusively at the edge, minimizing the risk of data leakage and ensuring the security of operations. Consequently, the data remains preserved.

2) *Distributed Processing*: The process of collecting crop data and training a model requires significant computational resources and time. However, FL offers a solution by enabling distributed processing, where training occurs simultaneously on multiple edge nodes. This approach helps to save computational resources and reduce the time required for training.

3) *Decentralized Data Storage*: In contrast to the traditional ML approach, FL enables crop data to be stored in a distributed manner across multiple devices or servers rather than being centralized in a single location. This decentralized data storage offers several advantages, including addressing privacy and security concerns. Since the data remains on the local devices or servers and is not transmitted to a central server, the risk of data breaches or unauthorized access to sensitive information is reduced. Additionally, FL leads to improved scalability and efficiency.

4) *Reduced Communication Overheads*: If the clients or local servers are located far away from the centralized server, transferring data from the local servers to the central server becomes challenging. This issue may arise if the local servers are in areas without communication lines or if there are frequent disruptions in the communication lines. Consequently, transferring data to the central server becomes problematic. To address this, decentralized data storage approach can be used, where the model trains at the local servers and then transmits the updated weights to the central server. This decentralized approach can significantly reduce the cost of data transmission. Additionally, to minimize upload times, the communication between the clients and the central server is kept to a minimum by only exchanging model updates, regardless of the size of the dataset. This approach helps to greatly reduce the communication costs involved.

5) *Improved Security*: The security of crop data is guaranteed as it is exclusively in the possession of the farmers.

6) *Energy Efficient*: Significant computational power is required when training the model on a central server. In FL, models are trained using smaller datasets, which results in lower energy and power consumption compared to traditional

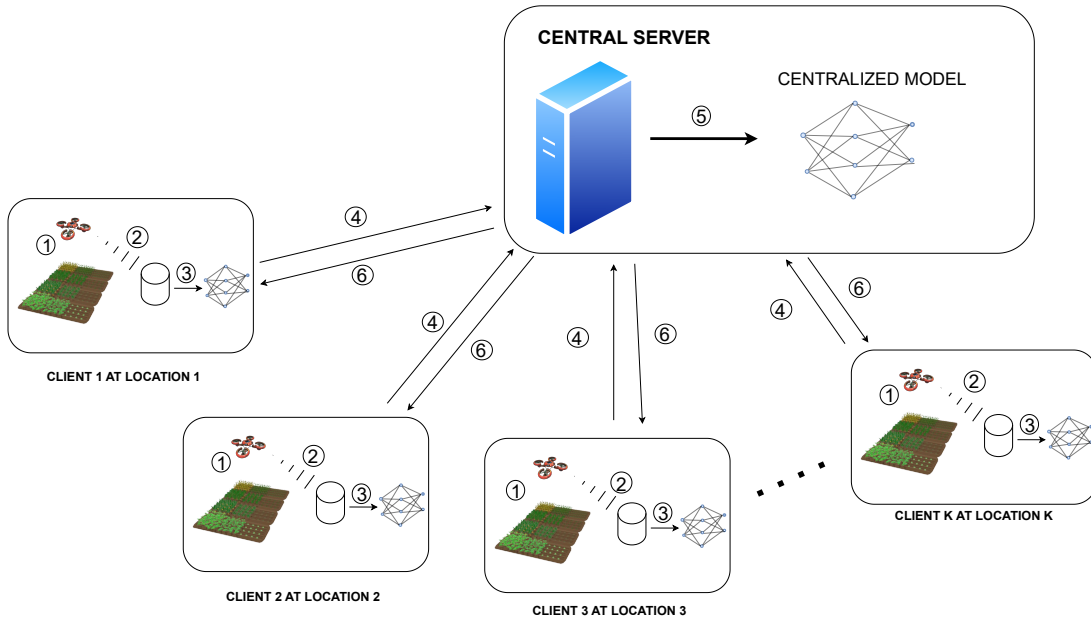


Fig. 2. The architecture of FL in smart agriculture

ML methods. As a result, devices with limited computational power, such as mobile phones, can effectively perform the model training.

7) *Continuous Learning*: In certain cases, the data that is received is in analog form rather than being discrete. Continuous learning facilitates the process of learning from the continuous flow of data, enabling adaptation to environmental changes and enhancing real-time performance.

8) *Collaborative Learning*: The approach known as collaborative learning involves training multiple classifier heads of the same network on the same training data simultaneously. This is done to enhance generalization and increase resistance to label noise, all without incurring any additional inference cost. Another term used to describe this approach is FL, as it involves using multiple sources to individually train the model and then combining them into a single model. This helps to improve accuracy and decrease the training time required for the model.

9) *Scalability*: Federated learning enables the storage of datasets in various locations and the concurrent training of models. This speeds up the deployment of AI and IoT models and enhances operational scalability spatially.

10) *Improved Data Diversity*: In the agricultural sector, FL algorithms are utilized to train models using data from various crops situated in different geographical regions. This approach leads to the creation of comprehensive models as the dataset consists of crops with diverse characteristics such as humidity, pest infestation, heavy metal content, and crop fertility. Consequently, the models developed through this method exhibit enhanced capacity to effectively handle new data, accommodate variations, and mitigate bias.

II. LITERATURE REVIEW

A. Disease Classification

Mehta et al. [9] used a lightweight convolutional neural network model to identify and categorize mango leaf diseases. The effectiveness of this lightweight model was evaluated in four different clients and five disease categories, including healthy, anthracnose, powdery mildew, leaf spot, and leaf curl. The researchers achieved a detection accuracy of 98% for these particular diseases. Table 1 presents a comparative analysis of existing studies.

In a study by Aggarwal et al. [6], the EfficientNetB3 model was used to detect rice diseases. The data set consisted of 5,932 images from rice fields in Odisha, India, as well as an agricultural and insect pest image database. The model achieved an accuracy of 99.79% for IID data and 97.48% for non-IID data.

The classification of pear disease was performed by Dockendorf et al. [17]. The authors integrated FL for training the various CNNs in TinyML devices and benchmarked different CNNs.

B. Disease Severity Assessment

Mehta et al. [11] used MobileNet to identify apple leaf disease and assess its severity levels. They examined the effectiveness of the model in four different categories of severity: healthy, mild, moderate, and severe. The focus was on two classes of specific diseases, namely apple scab, marsonina coronaria, and Alternaria leaf blotch. The researchers achieved a remarkable accuracy of 99.2% in detecting these diseases.

C. Crop Classification

Idoje et al [8] utilized Hyper tuned Federated averaging models for crop classification. The main objective of the study

TABLE I
COMPARATIVE ANALYSIS OF EXISTING STUDIES

Reference	Type of problem	Models	Accuracy	Number of clients
[9]	Disease detection	Lightweighted CNN	98%	4
[10]	Disease detection	Vector machines, K-nearest neighbors, Random forest, Logistic regression, MobileNet	99.20%	-
[11]	Disease detection	Vector machines, Linear,quadratic, Cubic regresion, Resnet	94.7	-
[12]	Disease detection	Resnet-101	90.27	6
[6]	Disease detection	EfficientNetB3	99.79	10
[8]	Crop classification	Naive bayes, Classifier chain	91(doubt)	-
[13]	Yield prediction	LASSO, Random forest, Resnet-16, Resnet-28	99.95	3
[14]	Intrusion detection system	FedACNN	99.76	-
[4]	Intrusion detection system	DNN,CNN and RNN	98.54	5,10,15
[15]	Intrusion detection system	IDSFNET	95.41	5,10,15
[16]	Pest detection with UAV	EfficientnetB3	99.55	-

was to predict chickpea crops in the presence of Rice and Maize crops. The results showed that the model achieved an accuracy of 91%.

D. UAV for Pest Detection

Khan et al. [16] employed FL-based unmanned aerial vehicles (UAVs) to classify various pests. They used an EfficientNet B03 for classification and detection purposes and compared the results with other models such as ANN, SVM, KNN and Naive Bayes. The CNN model demonstrated the highest accuracy of 99.55%.

Deng et al. [12] utilized a faster R-CNN to detect orchard pests. They employed the Online Hard Example Mining (OHEM) network training method to improve detection accuracy. The experimental findings indicated that the improved model achieved an accuracy of 90.27% for the detection of multiple pests.

E. Intrusion Detection System

Man et al [14] proposed an intelligent intrusion detection system using edge-assisted IoT. They utilized the Federated Attention Convolution Neural Network (FedACNN) and evaluated their model using the NSL-KDD dataset. The model achieved an accuracy of 99.76%. Compared to the Federated Average (FedAVG) model, the number of communication rounds was reduced to 50%.

Friha et al. [4] conducted a study on the performance of the Intrusion Detection System (IDS) using three different data sets. CSE-CIC-IDS2018, MQTTset, and InSDN. They experimented with three deep learning classifiers: deep neural networks (DNN), recurrent neural networks (RNN), and convolutional neural networks (CNN). The FL-based Intrusion Detection System (FELIDS) using a DNN model achieved an accuracy of 98.54% with the dataset collected from InSDN, outperforming other systems..

Zainuddin et al. [15] developed and evaluated the performance of a lightweight DNN model called IDSFedNet on the InSDN and Edge-IIoTset datasets. Their framework

utilized the Chi-square and Pearson correlation coefficients to reduce the complexity of the model and improve performance. The model outperformed existing FL-based intrusion detection models, achieving accuracies of 90.83% for the InSDN dataset and 95.41% for the Edge-IIoTset dataset.

F. Yield Forecasting

Manoj et al. [13] used a deep residual network to predict the soybean yield. They evaluated the performance of this model by comparing it with other learning models, namely LASSO, Random Forest, and Resnet-28. The results obtained demonstrated an accuracy of 99.95%.

G. Crop Risk Analysis

Yu et al. [18] introduced a risk-entropy model that uses FL. This model assesses the risk associated with crops, considering various factors such as policy, economic conditions, knowledge and information, infrastructure, cultural aspects, access to food, climate conditions, and social challenges. This model quantitatively describes the level of risk for agricultural products. They tested this model using a dataset of 90,510 agricultural product samples from 2015 to 2019. To implement this model in smart agriculture, three components are necessary: parties, a manager (server), and a framework. Parties refer to the data owners who possess the data, while the manager is a device that aggregates local models, performs the required computations, and sends back the updated model. To facilitate computation, privacy mechanisms, scheduling, and frameworks are essential.

TOOLS

There are several tools available to implement FL. Some popular FL tools include TensorFlow Federated (TFF) [19], PySyft [20], IBM Federated Learning Framework [21], [22], and Fate framework [23]. These tools provide the libraries necessary to enable FL in various agricultural applications.

H. Matlab

Matlab serves as a valuable tool for analyzing and designing systems and products. The use of Matlab for model simulation is widespread among researchers [24], [25]. It offers a wide range of libraries for the execution and construction of machine learning algorithms and neural networks. Matlab is compatible with most models.

I. TensorFlow Federated(TFF)

Google's TensorFlow offers an API that allows users to develop and customize their own models for experimentation. TensorFlow Federated (TFF) is an open source framework that extends Keras models to enable computation on vertically and horizontally distributed data. With TFF, developers can apply FL algorithms to their models and data. Keras, on the other hand, provides a user-friendly and efficient interface to solve machine learning problems.

J. Fate

The framework Fate [26] enables secure computation through the use of homomorphic encryption and multiparty computation, making it widely recognized and popular. It offers support for a range of machine learning algorithms, including logistic regression, tree-based algorithms, and deep learning techniques such as deep learning neural networks and transfer learning. Over time, Fate has evolved into various branches, each serving different purposes. These branches provide a wide array of services, with FATE-cloud offering industrial-level Federated Learning cloud services, the Eggroll framework facilitating the implementation of machine learning models, and the FATE-LLM framework specifically supporting FL for large language models (LLMs).

K. IBM Federated learning

IBM Federated Learning, which was created by IBM Technologies [27], includes supervised, unsupervised, and reinforcement learning. It can work with any neural network topology that is compatible with Keras, PyTorch, and TensorFlow. Furthermore, it supports Decision Tree ID3, linear classifiers, linear regression, linear SVM, ridge regression, k-means clustering, and Nave Bayes. It also supports deep reinforcement learning models such as DQN, DPPG, PDO, and others.

L. Flower Framework

Researchers widely adopt the Flower framework because of its versatility. It offers both an Android SDK and an iOS SDK, making it compatible with a wide range of devices. Additionally, Flower can seamlessly integrate with various other frameworks, including PyTorch, TensorFlow, Hugging Face Transformers, Scikit-Learn, and more.

M. Pysyft

PySyft is a widely adopted framework that employs various techniques, such as federated learning, differential privacy, and encrypted computation, to separate private data from model training. Currently, it exclusively supports the deep learning libraries PyTorch and TensorFlow. However, starting with version 0.7, it will also offer a TensorFlow Federated API.

We observe the properties of other tools like NVFLARE, OpenFL, Fedml, Fedlab, FedJAX, FedTree, etc. in the table II.

III. DISCUSSIONS

FL enables the devices to collaboratively learn from the model while keeping the data to themselves, unlike traditional learning, where the data is transferred to the central server. The main reason for the need for FL is farmers' concern about the privacy of their crop data. According to the survey conducted by the American Farm Bureau Federation, 77% of farmers are concerned about the privacy of their data [36].

We need hardware and software elements to implement Federated learning. In the hardware segment, there are three essential components: end devices, training devices, and databases. Environmental sensors are responsible for monitoring various factors such as soil conditions, humidity, plant health, and temperature. Soil sensors are specifically designed to monitor pH and metal concentration. While sensors collect data, devices process and analyze it based on criteria such as the number of features and their complexity. To ensure privacy, additional techniques such as privacy encoding and homomorphic encryption can be used.

For low-complexity models with simple characteristics, microcontrollers such as Arduino, Raspberry Pi, and mobile phones can be utilized. However, clients involved in decentralized federated learning require more computing power to communicate and share model parameters without relying on a central server. Training on mobile phones can be challenging. Since the model is trained locally, it can be stored on mobile phones or any SSDs connected to microcontrollers.

When it comes to software requirements, we require a framework that can run our model, manage the aggregation process, and handle a large number of devices, ranging from thousands to millions. We have previously discussed several frameworks in the previous sections.

Although Federated Learning shows great potential, there exist various challenges and limitations in its implementation, as outlined in the following sections.

A. Communication

Areas with limited network bandwidth and heterogeneous networks, such as remote rural areas where agriculture is the main industry, are expected to experience increased communication costs, an increased number of communication rounds, and reduced latency. However, there is a lack of techniques that efficiently support communication by reducing the total number of communication rounds and operating effectively in areas of high bandwidth [37].

TABLE II
COMPARATIVE ANALYSIS OF FEDERATED LEARNING TOOLS

Tools	Supporting languages	Other Supporting Frameworks	Aggregation Algorithms	Participation type
NVFLARE [28]	Python	PyTorch, TensorFlow, Scikit-learn, XGBoost	FedProx, FedOpt, SCAFFOLD, Ditto	Cross-silo
OPEN FL [29]	Haxe, JavaScript , TypeScript	TensorFlow and PyTorch	FedProx, FedAvg, FedOpt, FedCurv	Cross-silo Cross-device
FED ML [30]	Python	TensorFlow, PyTorch, MXNet	FedAvg, FedOpt, FedNova	Cross-silo
FED LAB [31]	Python	TensorFlow, PyTorch, MXNet	FedAvg, FedAsgd	Cross-silo Cross-device
FedJAX [32]	Python	Any framework be defined using the JAX syntax	FedAvg, krum, Mean, Weighted FedAvg	Cross-device
FedTree [33]	Python	XGBoost	FedAvg	Cross-silo
TFF [34]	Python, Javascript, Java, C++	any compatible framework	FedAvg, Any user defined algorithm	cross-silo
FLOWER [35]	Python	TensorFlow, PyTorch, Keras	FedAdagrad, FedYogi, FedOpt, FedAdam	cross-silo
PYSYFT [20]	Python	PyTorch and TensorFlow	FedAvg	Cross-silo Cross-device

B. Energy-efficient routing

Energy-efficient routing in Federated Learning faces challenges like limited battery life of devices, wireless communication overhead, heterogeneous device capabilities, dynamic network topologies. Addressing these issues requires a combination of techniques, such as energy-aware routing protocols [38], [39], efficient communication schemes, resource allocation strategies, and energy-efficient machine learning algorithms tailored for FL environments.

C. Privacy Concerns

When sharing model updates privacy concerns may arise for farmers, as this may involve sharing of sensitive information such as crop yields or soil conditions. Simply providing gradient information can still compromise privacy to external parties.

D. Data Heterogeneity

Crops may have experienced different environmental conditions, resulting in variations in their yield. Data collected by the devices and sensors in the farms exhibit a nonidentical and nonindependent distribution. This data generation process poses challenges in determining the accuracy and reliability of FL models. Applying the traditional approach of direct average aggregation can negatively impact the convergence and precision of the model.

E. System Heterogeneity

Farms commonly utilize a combination of devices with varying hardware capabilities, operating systems, and software versions. This can result in compatibility issues and errors when uploading models, which can limit the performance of the unified model. Additionally, network size and system-related constraints often mean that only a small fraction of devices are active at any given time.

F. Limited Control over Data Quality

The task of ensuring the accuracy of agricultural data can be difficult because of the unpredictable changes in the environment and the limited control over the local training data. This lack of data quality will have an impact on the model.

G. Limited Debugging and Analysis

The data in agriculture is typically diverse, with variations observed across farms due to factors such as soil types, climates, and farming practices. Additionally, the data changes quickly due to weather conditions and seasonal variations. Consequently, it becomes challenging to monitor the performance of the model over time and pinpoint specific issues.

IV. CONCLUSIONS AND FUTURE SCOPE

Federated Learning is a rapidly growing research field, although still in its early stages despite significant contributions. This article explores the benefits of FL, the challenges it faces in implementation, and its limitations. There is ample opportunity for further progress and improvements, especially in the context of smart agriculture. Although privacy concerns have been addressed with respect to training data, potential threats could impact model aggregation. To address privacy issues in FL, methods such as homomorphic encryption could be utilized. Future focus should prioritize the development of algorithms to maintain privacy across edge devices and data tiers. Integrating next-generation wireless systems, such as 6G networks, with FL-enabled Industrial Internet of Things (IIoT) networks could improve privacy protection. The use of edge nodes for FL and advancements such as FedACNN for edge-assisted IoT security are promising. Research on intrusion detection in edge-assisted IoT traffic data is crucial. Exploring CNN architectures, such as the Swin Transformer, for agricultural applications such as crop variety prediction is essential. More research is needed for the classification of diseases and other aspects of FL in agriculture. Improving FL

algorithms and convergence speed, as well as identifying dense targets, are vital to improve model accuracy and performance.

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